Activity 1.6 – Priority Encoder and Decoder

1. Design Approach:

Designed an 8-to-3 priority encoder and a 3-to-8 decoder using Verilog HDL. The priority encoder is implemented using two methods: case-based logic and if-else logic. The highest-priority input is encoded into a 3-bit output.

2. Verification Steps:

- Created a separate testbench.
- Tested individual encoder input combinations.
- Checked priority when multiple inputs are active.
- Compared both encoder implementations.
- Tested all eight decoder output combinations.
- Simulated using ModelSim.

3. Results:

The testbench checks encoder outputs, valid signals and decoder outputs. PASS/FAIL messages are displayed.

4. Issues Faced:

Priority must be maintained when multiple inputs are active. Default values are required to avoid unwanted latch inference**.**

5. Conclusion:

Successfully designed and verified an 8-to-3 priority encoder and a 3-to-8 decoder.